2.5 Testing

Testing is a vital process in software development used to analyze and verify that a program runs as expected, satisfies user requirements, and performs reliably under diverse conditions. It entails executing the software with precise inputs and analyzing the outputs to find flaws, logical mistakes, or variations from expected behavior. A test case is an organized set of conditions, inputs, and expected results used during testing to assess whether a particular feature or function of the system is performing appropriately. Test cases help guarantee that each component of the software fulfills its intended role under both regular and edge-case scenarios.

Testing is vital because it enhances the quality, usability, and stability of the software, letting developers detect errors early in the development cycle before the system is deployed. It decreases the possibility of system crashes, incorrect outputs, and security vulnerabilities, all of which can lead to user displeasure or operational failures. Effective testing guarantees that the software operates reliably and helps create faith in the system's stability.

In general, the benefits of testing include increased system correctness, smoother user experience, cost savings by preventing future faults, and enhanced maintainability. Testing also promotes easier debugging and code rewriting, since developers may confidently make changes knowing that automated or repetitive test cases will discover any regressions. In the context of the Doctor Channelling System, testing plays a critical role in ensuring that essential operations like patient registration, doctor scheduling, and appointment management perform correctly and handle invalid inputs gracefully. For example, testing prohibits users from inputting non-numeric values for age or charge, ensures that duplicate submissions are not accepted, and checks that the waiting list system operates as expected when slots are full. By methodically evaluating each component, the system becomes more robust, user-friendly, and dependable in real-world healthcare scheduling scenarios, where accuracy and confidence are vital.

2.5.1 Test Plan and Test Cases

Test Plan

Test ID	Test Name	Expected Outcome
DCSO1	Register patient with Empty ID	An Error Message “Error: Patient ID cannot be empty”
DCS02	Register patient with Duplicate ID	An Error Message “Error: Patient ID already exists”
DCS03	Register Patient with Non-numeric age	An Error Message “Error: Age must be a valid number”
DCS04	Search an unregistered patient	An Error Message “Error: Patient not found”
DCS05	Register Doctor with empty name	An Error Message “Error: Doctor name cannot be empty”
DCSO6	Register Doctor with a Negative fee	An Error Message “Error: Fee must be a Postive value”
DCS07	Search for an unregistered doctor	An Error Message “Error: Doctor not found”
DCS08	Book an appointment with empty Patient ID	An Error Message “Error: Patient ID cannot be empty”
DCS09	Book an appointment with empty Doctor ID	An Error Message “Error: Doctor ID cannot be empty”
DCS10	Cancel an appointment of a Patient with no Appointments	An Error Message “Error: Appointment not found for the given Patient ID”
DCS11	Reschedule a Patient with no Appointments	An Error Message “Error: No active appointment found for this patient”
DCS12	Book an Appointment for a Doctor who’s slots are full	A Message “All slots are full. Patient added to waiting list”

DCS13	Cancel an Appointment of a Doctor who's slot is full (Waiting list Implementation)	A Message "Appointment cancelled for (Cancelled patient name)", "Waiting Patient (Waiting patient name) assigned to freed slot: slot number and time"
DCS14	Choose an invalid number in the menu	An Error Message "Invalid choice, Please enter a number between 1 and 10"
DCS15	View Patient List- When Empty	A Message "No Patients Registered"
DCS16	View Doctor List- When Empty	A Message "No Doctors registered"
DCS17	View Appointment List- When Empty	A Message "No appointments currently

Test Cases

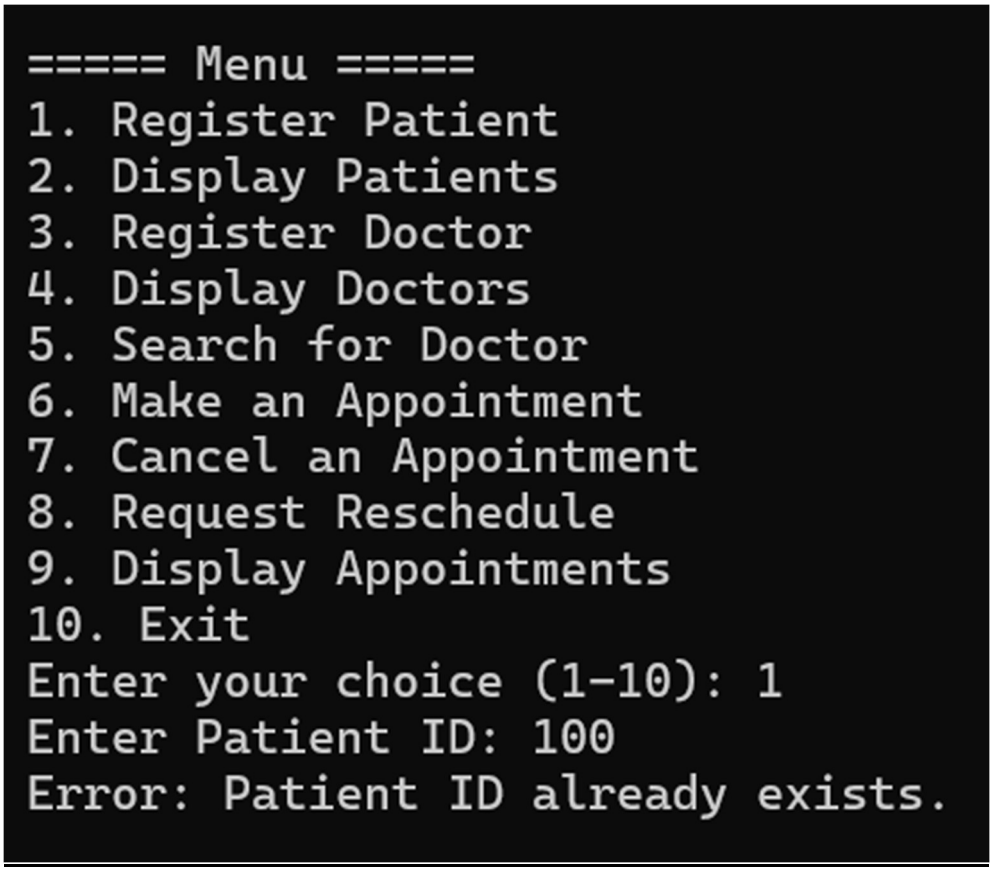
Test ID	DBS01
Test Name	Register patient with Empty ID
Expected Results	An Error Message "Error: Patient ID cannot be empty"
Actual Results	

```
--- Doctor Channelling System ---

===== Menu =====
1. Register Patient
2. Display Patients
3. Register Doctor
4. Display Doctors
5. Search for Doctor
6. Make an Appointment
7. Cancel an Appointment
8. Request Reschedule
9. Display Appointments
10. Exit
Enter your choice (1-10): 1
Enter Patient ID:
Error: Patient ID cannot be empty.
```

Status	Successfully displayed an error message, Test passed!

Test ID	DBS02
Test Name	Register patient with Duplicate ID
Expected Results	An Error Message “Error: Patient ID already exists”
Actual Results	



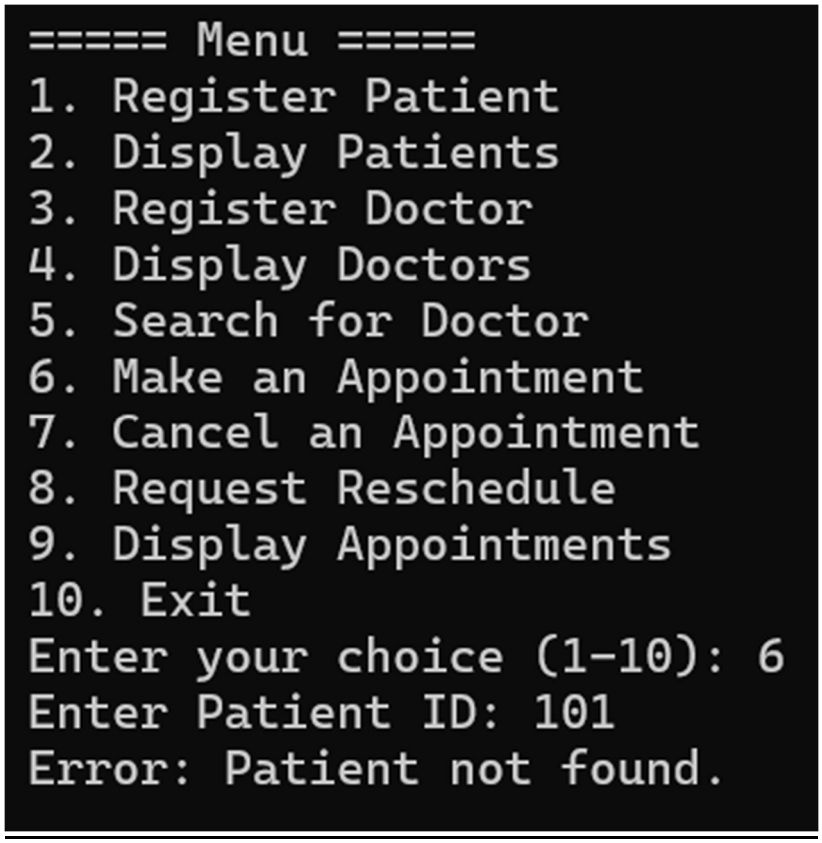
Status	Successfully displayed an error message, Test passed!

Test ID	DBS03
Test Name	Register Patient with Non-numeric age
Expected Results	An Error Message “Error: Age must be a valid number”
Actual Results	

```
===== Menu =====
1. Register Patient
2. Display Patients
3. Register Doctor
4. Display Doctors
5. Search for Doctor
6. Make an Appointment
7. Cancel an Appointment
8. Request Reschedule
9. Display Appointments
10. Exit
Enter your choice (1-10): 1
Enter Patient ID: 1
Enter Patient Name: Kavindhya
Enter Mobile Number: 9087563278
Enter Email: Kavi@gmail.com
Enter City: Kandy
Enter Age: Twentyfour
Error: Age must be a valid number.
```

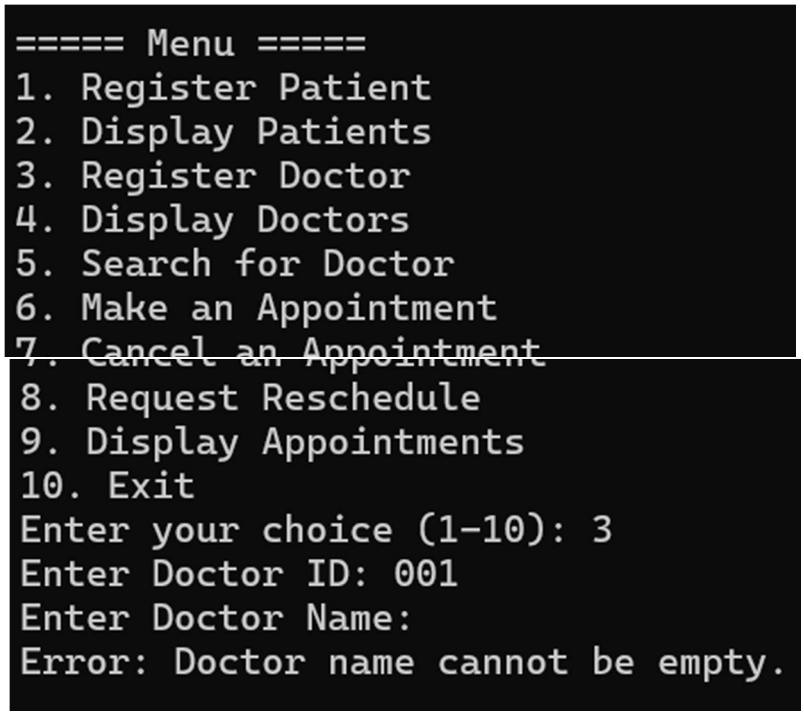
Status	Successfully displayed an error message, Test passed!

Test ID	DBS04
Test Name	Search an un-registered Patient
Expected Results	An Error Message “Error: Patient not found”
Actual Results	



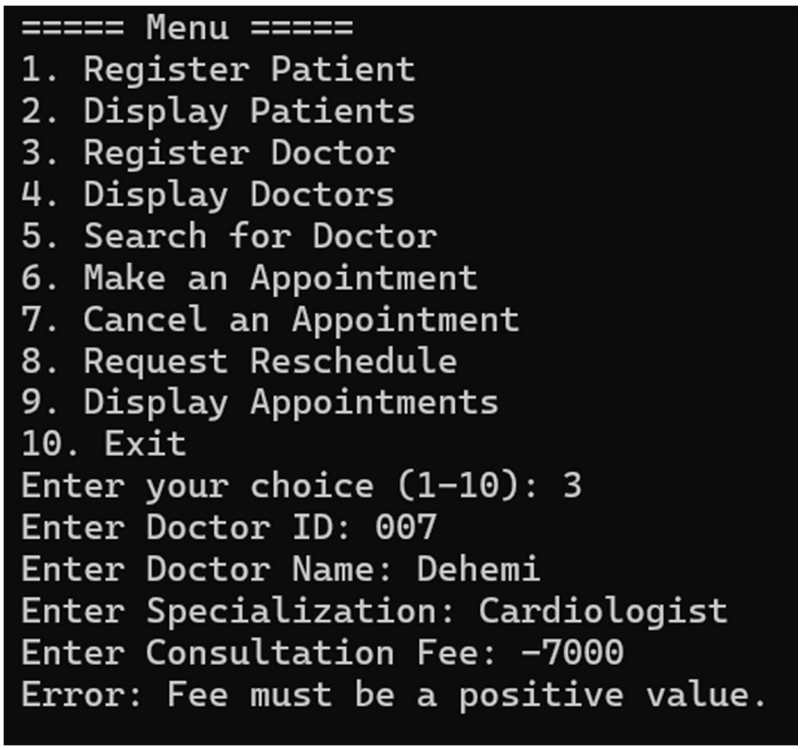
Status	Successfully displayed an error message, Test passed!

Test ID	DBS05
Test Name	Register Doctor with empty name
Expected Results	An Error Message “Error: Doctor name cannot be empty”
Actual Results	



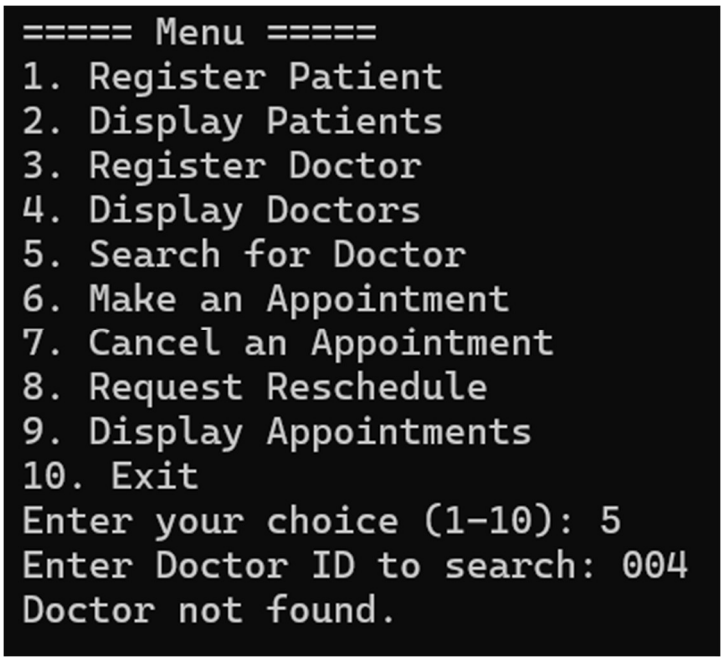
Status	Successfully displayed an error message, Test passed!

Test ID	DBS06
Test Name	Register Doctor with a Negative fee
Expected Results	An Error Message “Error: Fee must be a Postive value”
Actual Results	



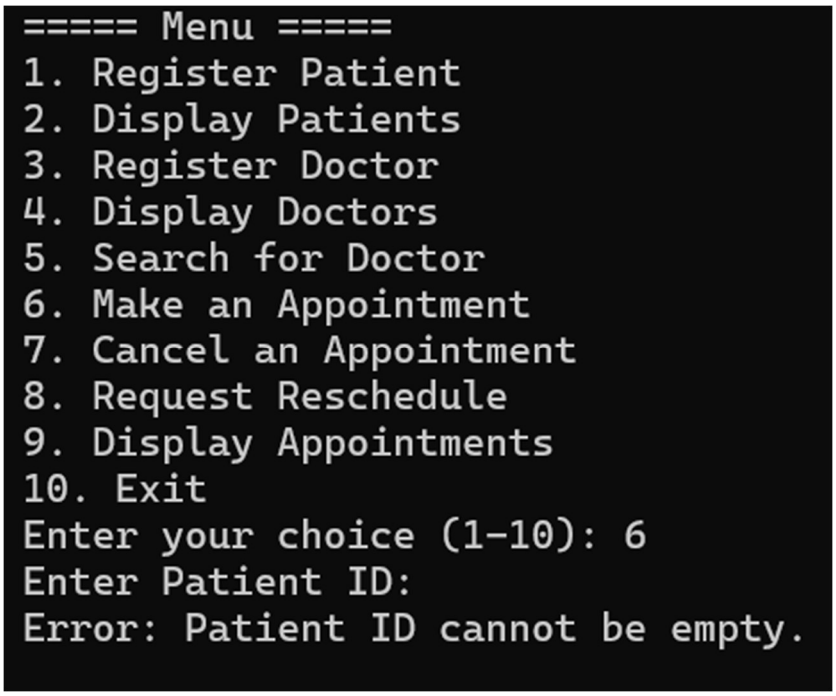
Status	Successfully displayed an error message, Test passed!

Test ID	DBS07
Test Name	Search for an unregistered doctor
Expected Results	An Error Message “Error: Doctor not found”
Actual Results	



Status	Successfully displayed an error message, Test passed!

Test ID	DBS08
Test Name	Book an appointment with empty Patient ID
Expected Results	An Error Message “Error: Patient ID cannot be empty”
Actual Results	



Status	Successfully displayed an error message, Test passed!

Test ID	DBS09
Test Name	Book an appointment with empty Doctor ID
Expected Results	An Error Message “Error: Doctor ID cannot be empty”
Actual Results	

```
===== Menu =====
1. Register Patient
2. Display Patients
3. Register Doctor
4. Display Doctors
5. Search for Doctor
6. Make an Appointment
7. Cancel an Appointment
8. Request Reschedule
9. Display Appointments
10. Exit
Enter your choice (1-10): 6
Enter Patient ID: 100
Enter Doctor ID:
Error: Doctor ID cannot be empty.
```

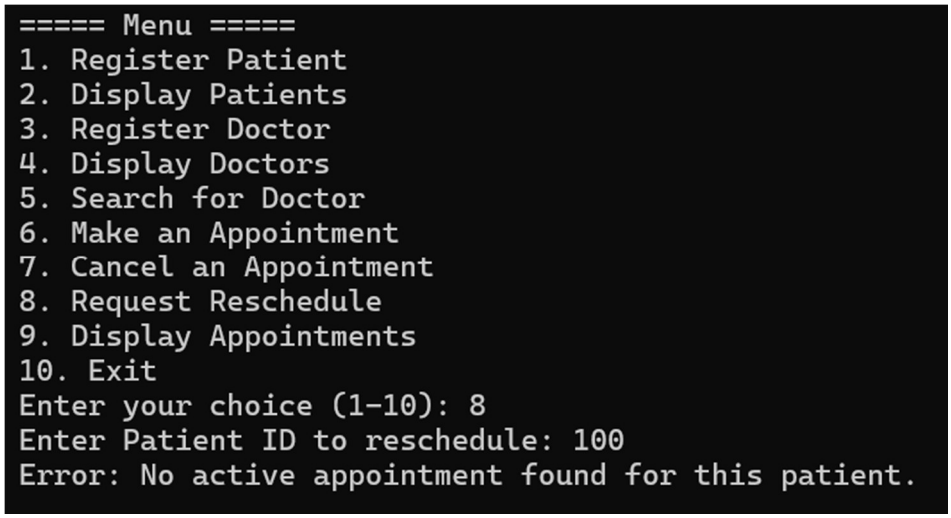
Status	Successfully displayed an error message, Test passed!

Test ID	DBS10
Test Name	Cancel an appointment of a Patient with no Appointments
Expected Results	An Error Message “Error: Appointment not found for the given Patient ID”
Actual Results	

```
===== Menu =====
1. Register Patient
2. Display Patients
3. Register Doctor
4. Display Doctors
5. Search for Doctor
6. Make an Appointment
7. Cancel an Appointment
8. Request Reschedule
9. Display Appointments
10. Exit
Enter your choice (1-10): 7
Enter Patient ID to cancel appointment: 100
Error: Appointment not found for the given Patient ID.
```

Status	Successfully displayed an error message, Test passed!

Test ID	DBS11
Test Name	Reschedule a Patient with no Appointments
Expected Results	An Error Message “Error: No active appointment found for this patient”
Actual Results	



Status	Successfully displayed an error message, Test passed!

Test ID	DBS12
Test Name	Book an Appointment for a Doctor who’s slots are full (Waiting list Implementation)
Expected Results	A Message “All slots are full. Patient added to waiting list”
Actual Results	

```
===== Menu =====
1. Register Patient
2. Display Patients
3. Register Doctor
4. Display Doctors
5. Search for Doctor
6. Make an Appointment
7. Cancel an Appointment
8. Request Reschedule
9. Display Appointments
10. Exit
Enter your choice (1-10): 4
Doctor ID: 001, Name: Nimesha, Specialization: General Physician, Fee: Rs. 3500.0
Available Slots:

===== Menu =====
1. Register Patient
2. Display Patients
3. Register Doctor
4. Display Doctors
5. Search for Doctor
6. Make an Appointment
7. Cancel an Appointment
8. Request Reschedule
9. Display Appointments
10. Exit
Enter your choice (1-10): 6
Enter Patient ID: 107
Enter Doctor ID: 001
All slots are full. Patient added to waiting list.
```

Status	Successfully displayed an error message, Test passed!

Test ID	DBS13
Test Name	Cancel an Appointment of a Doctor who's slot is full (Waiting list Implementation)
Expected Results	A Message "Appointment cancelled for (Cancelled patient name)", "Waiting Patient (Waiting patient name) assigned to freed slot: slot number and time"
Actual Results	

```
===== Menu =====
1. Register Patient
2. Display Patients
3. Register Doctor
4. Display Doctors
5. Search for Doctor
6. Make an Appointment
7. Cancel an Appointment
8. Request Reschedule
9. Display Appointments
10. Exit
Enter your choice (1-10): 7
Enter Patient ID to cancel appointment: 106
Appointment cancelled for Preethi
Waiting patient Lakshman assigned to freed slot: No6:11:30am
```

Status	Successfully displayed an error message, Test passed!

Test ID	DBS14
Test Name	Choose an invalid number in the menu
Expected Results	An Error Message “Invalid choice, Please enter a number between 1 and 10”
Actual Results	

```
===== Menu =====
1. Register Patient
2. Display Patients
3. Register Doctor
4. Display Doctors
5. Search for Doctor
6. Make an Appointment
7. Cancel an Appointment
8. Request Reschedule
9. Display Appointments
10. Exit
Enter your choice (1-10): 12
Invalid choice. Please enter a number between 1 and 10.
```

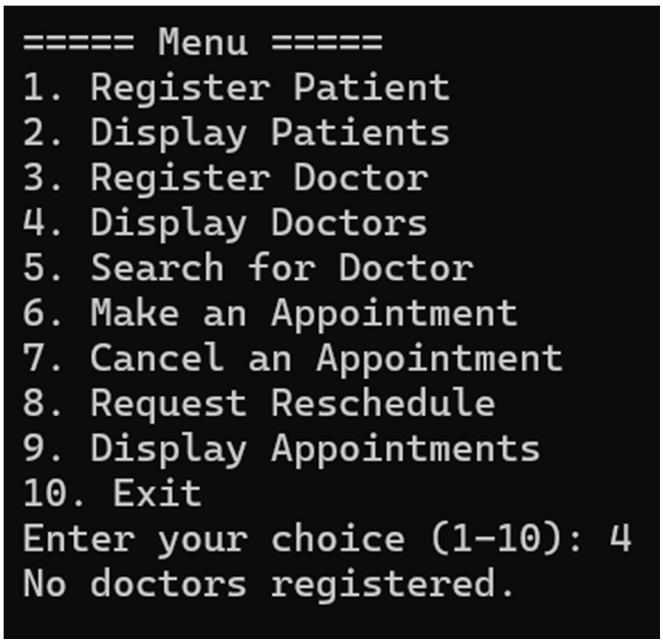
Status	Successfully displayed an error message, Test passed!

Test ID	DBS15
Test Name	View Patient List- When Empty
Expected Results	A Message “No Patients Registered”
Actual Results	

```
===== Menu =====
1. Register Patient
2. Display Patients
3. Register Doctor
4. Display Doctors
5. Search for Doctor
6. Make an Appointment
7. Cancel an Appointment
8. Request Reschedule
9. Display Appointments
10. Exit
Enter your choice (1-10): 2
No patients registered.
```

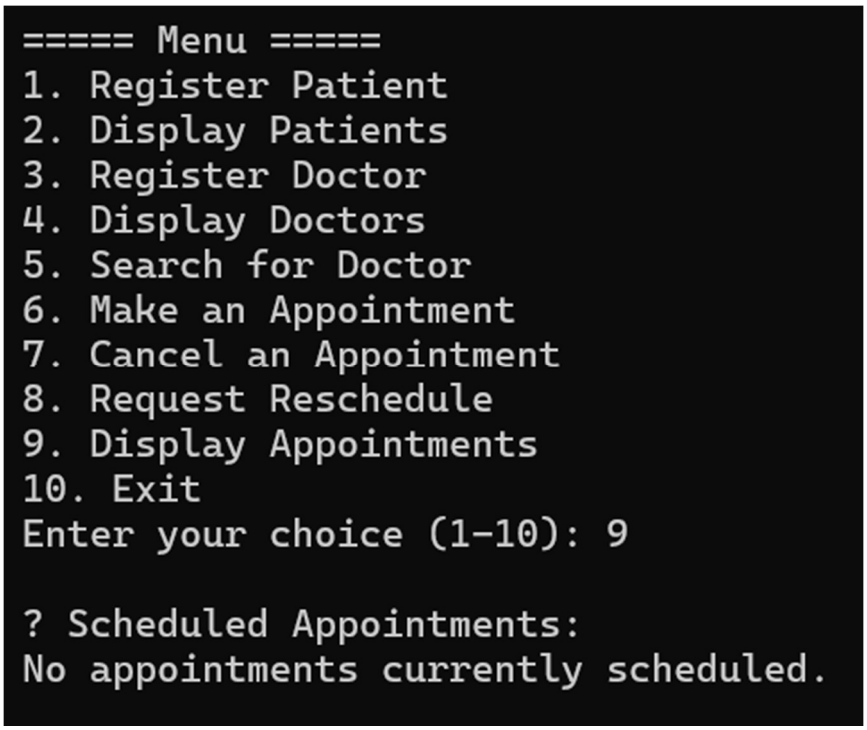
Status	Successfully displayed an error message, Test passed!

Test ID	DBS16
Test Name	View Doctor List- When Empty
Expected Results	A Message “No Doctors registered”
Actual Results	



Status	Successfully displayed an error message, Test passed!

Test ID	DBS17
Test Name	View Appointment List- When Empty
Expected Results	A Message “No appointments currently scheduled”
Actual Results	



Status	Successfully displayed an error message, Test passed!